Amlogic Hybrid Android R Release Note

AMLOGIC, Inc.
2518 Mission College Blvd
Santa Clara, CA 95054
U.S.A.
www.amlogic.com

Content

1. Basic Information

1.1. Introduction

This document is the release notes of Amlogic reference source codes aligned with Android R running on Amlogic S905X4/S905C2/S905Y4/S905C3/S805X2 reference hardware.

S905X4/S905C2/S905Y4/S905C3 released the GTVS clean version(Hybrid GM).

S805X2 released the GTVS clean version(OTT GM).

The purpose of this release is:

1. GTVS Clean.
   (1) CTS(11.0 R5)
   (2) GTS(9.0 r1)
   (3) VTS(11.0 R5)
   (4) CTS-verify(11.0 R5)
   (5) CTS-on-gsi(11.0 R5)
   (6) STS(2021-10)
   (7) TVTS(2.0 R1)
   (8) Android TV smoking(Youtube testing)
2. Pre-certification
   (1) HDMI CTS 2.0 pass
   (2) HDMI CEC pass
   (3) HDR10+ pass
   (4) HDCP 1.4/2.3 pass
   (5) VMX-IPTV Client E2E
   (6) VMX-WEB Client E2E
   (7) VMX-DVB Client E2E
   (8) Dolby vision v2.5 pass
   (9) DDP V4.8 for DVB pass
   (10) MS12 V2.4 for DVB pass
   (11) NTS pre-cert
   (12) ART
3. Full function ready:
   (1) Component compatibility
      (1) HDMI
      (2) CVBS
      (3) WIFI
      (4) BT
      (5) Ethernet
      (6) Bluetooth
      (7) USB
      (8) Remote
      (9) Spdif
   (2) Suspend/Resume
   (3) Multi-media functionality.
   (4) Multi-instance.
   (5) Widevine CAS.
   (6) Media CAS V2.
   (7) Display functionality.
   (8) PQ/DI.
   (9) HDR10.
   (10) Dolby Vision.

(11) System functionality.
(12) Audio-output.
(13) Secure boot V3.
(14) Fastboot
(15) A/B update
(16) W1 WIFI(v1.2.13_20210906)
(17) Nagra TKL
(18) Nagra Nocs

4. Performance
    (1) Benchmark
    (2) UX
    (3) APP launch speed
    (4) Video start/Switch speed
    (5) Memory usage
5. Stability.
    (1) Auto reboot - 72h
    (2) Auto Shutdown - 72h
    (3) GPU - 3D Game - 72h
    (4) Monkey test - 72h
    (5) Factory reset - 72h
    (6) A/V Burn - 72h
    (7) High/Low temperature - 72h
6. DTVKit
    (1) DVB-C Program search
    (2) DVB-S/DVB-S2 Program search
    (3) DVB-T/DVB-T2 Program search
    (4) Channel auto search
    (5) Channel manual search
    (6) NIT search
    (7) MHEG5
    (8) Teletext
    (9) PVR
    (10) TimeShift
    (11) subtitle
    (12) Program Edit
    (13) LCN
    (14) EPG
    (15) Channels Lock
    (16) Parental Lock
    (17) AD
    (18) Info Banner
    (19) Aspect Ratio
    (20) update channel
    (21) Date、Time
    (22) Audio language
    (23) Switch Channel
    (24) FCC/PIP functionality

1.2. Chip Information

S905X4/S905C2/S905Y4/S905C3/S805X2 :

We choose S905X4/S905Y4 reference platform as our major developing and testing platforms for this release.

Item	S905X4/S905C2	S805X2
CPU	Quad-core ARM Cortex-A55	Quad-core ARM Cortex-A35
GPU	ARM Mali-G31 MP2	ARM Mali-G31 MP2
Memory	32-bit DDR3/4/3L, LPDDR3/4	32-bit DDR3/4/3L, LPDDR4
Video decoding	AV1/H.265/VP9/AVS2-P2 to 4k@60fps	AV1/H.265/VP9/AVS2-P2 to 1080p@60fps
Video Encoding	H.265/JPEG to 1080p@60fps	SW encoder
HDMI-TX	4k60hz	1080p60hz
AV output	CVBS	CVBS
Ethernet	10M/100M/1000M	10M/100M
IP License	Dolby Audio/DTS/Dolby Vision	Dolby Audio/DTS
Demux	Support	Support

Item	S905Y4	S905C3
CPU	Quad-core ARM Cortex-A35	Quad-core ARM Cortex-A35
GPU	ARM Mali-G31 MP2	ARM Mali-G31 MP2
Memory	32-bit DDR3/4/3L, LPDDR4	32-bit DDR3/4/3L, LPDDR4
Video decoding	AV1/H.265/VP9/AVS2-P2 to 4k@60fps	AV1/H.265/VP9/AVS2-P2 to 4k@60fps
Video Encoding	SW encoder	SW encoder
HDMI-TX	4k60hz	4k60hz
AV output	CVBS	CVBS
Ethernet	10M/100M	10M/100M
IP License	Dolby Audio/DTS/Dolby Vision	Dolby Audio/DTS/Dolby Vision
Demux	Support	Support

1.3. Reference Platform

Mbox:

- **Ohm(S905X4)**
  EMMC,WIFI QCA6174 ,DDR4 2GB
- **Ohmcas(S905C2)**
  EMMC,WIFI QCA6174 ,DDR4 2GB
- **Oppen(S905Y4)**
  EMMC,WIFI QCA6174 ,DDR4 2GB
- **Oppencas(S905C3)**
  EMMC,WIFI QCA6174 ,DDR4 2GB
- **Planck(S805X2)**
  EMMC,WIFI QCA6174 ,DDR4 1GB

1.4. How to Get Code

You can download Android R source code by running the following repo commands:

1. use below command to get the code in google server and amlogic server(apply the license from Google and Amlogic, and apply the separate license for playready and tdk from Amlogic):

```
$cd~/<your-amlogic-repo-dir>/
$repo init -u ssh://git@openlinux.amlogic.com/r-amlogic/platform/manifest.git -b r-amlogic -m
openlinux/ott/r-amlogic-20211122-gtvs.xml
$repo sync
```

Notice: Oversea customer use below command:

```
$cd~/<your-amlogic-repo-dir>/
$repo init -u ssh://git@openlinux2.amlogic.com/r-amlogic/platform/manifest.git -b r-amlogic
-m openlinux/ott/r-amlogic-20211122-gtvs.xml
$repo sync
```

Needed storage space: need about 200G storage space.

1.5. How to config audio policy

According to your licensed audio technology, you should select the correct audio define in your build to select the correct audio_policy_configuration.xml.
For example, if you are using oppen platform, then you can find such define in /device/amlogic/oppen/oppen.mk

```
TARGET_DOLBY_MS12_VERSION := 2
TARGET_BUILD_DOLBY_DDP := true
TARGET_BUILD_DTSHD := true
```

The audio technology and define should match each other.

Audio technology	Yes	No
Dolby DDP	TARGET_BUILD_DOLBY_DDP := true	TARGET_BUILD_DOLBY_DDP := false
DTS HD	TARGET_BUILD_DTSHD := true	TARGET_BUILD_DTSHD := false
MS12 V2	TARGET_DOLBY_MS12_VERSION := 2	remove the define

If you select the Dolby MS12 v2 version, you don't need enable ddp again, it is already included in Dolby ms12.

1.6. How to config dolby vision in S905Y4-J/K

S905Y4-J/K supports dolby vision. You need patch the config as follows:

```
diff --git a/files/media_codecs_performance.xml b/files/media_codecs_performance.xml
index 5863ec5..4466595 100644
--- a/files/media_codecs_performance.xml
+++ b/files/media_codecs_performance.xml
@@ -58,6 +58,24 @@
         </MediaCodec>
     </Encoders>
     <Decoders>
+        <MediaCodec name="OMX.amlogic.dolby-vision.dvhe.decoder.awesome2"
type="video/dolby-vision"   update="true">
+            <Limit name="performance-point-1920x1080" range="120-120" />
+        </MediaCodec>
+        <MediaCodec name="OMX.amlogic.dolby-vision.dvhe.decoder.awesome2.secure"
type="video/dolby-vision"   update="true">
+            <Limit name="performance-point-1920x1080" range="120-120" />
+        </MediaCodec>
+        <MediaCodec name="OMX.amlogic.dolby-vision.dvav.decoder.awesome2"
type="video/dolby-vision"   update="true">
+            <Limit name="performance-point-1920x1080" range="120-120" />
+        </MediaCodec>
+        <MediaCodec name="OMX.amlogic.dolby-vision.dvav.decoder.awesome2.secure"
type="video/dolby-vision"   update="true">
+            <Limit name="performance-point-1920x1080" range="120-120" />
+        </MediaCodec>
+        <MediaCodec name="OMX.amlogic.dolby-vision.dav1.decoder.awesome2"
type="video/dolby-vision"   update="true">
+            <Limit name="performance-point-1920x1080" range="120-120" />
+        </MediaCodec>
+        <MediaCodec name="OMX.amlogic.dolby-vision.dav1.decoder.awesome2.secure"
type="video/dolby-vision"   update="true">
+            <Limit name="performance-point-1920x1080" range="120-120" />
+        </MediaCodec>
         <MediaCodec name="OMX.amlogic.avc.decoder.awesome2" type="video/avc" update="true">
             <Limit name="measured-frame-rate-320x240" range="653-653" />
             <Limit name="measured-frame-rate-720x480" range="440-440" />
```

```
--- a/files/media_codecs.xml
+++ b/files/media_codecs.xml
@@ -77,6 +77,69 @@ Only the three quirks included above are recognized at this point:

 <MediaCodecs>
     <Decoders>
+        <MediaCodec name="OMX.amlogic.dolby-vision.dvhe.decoder.awesome2" type="video/dolby-vision" >
+            <Limit name="size" min="64x64" max="3840x2160" />
+            <Limit name="alignment" value="2x2" />
+            <Limit name="block-size" value="16x16" />
+            <Limit name="blocks-per-second" min="1" max="1944000" />
+            <Limit name="bitrate" range="1-300000000" />
+            <Feature name="adaptive-playback" />
+            <Feature name="tunneled-playback" />
+            <Limit name="concurrent-instances" max="1" />
+        </MediaCodec>
+        <MediaCodec name="OMX.amlogic.dolby-vision.dvhe.decoder.awesome2.secure" type="video/dolby-vision" >
+            <Limit name="size" min="64x64" max="3840x2160" />
+            <Limit name="alignment" value="2x2" />
+            <Limit name="block-size" value="16x16" />
+            <Limit name="blocks-per-second" min="1" max="1944000" />
+            <Limit name="bitrate" range="1-300000000" />
+            <Feature name="adaptive-playback" />
+            <Feature name="secure-playback" required="true" />
+            <Feature name="tunneled-playback" />
+            <Limit name="concurrent-instances" max="1" />
+        </MediaCodec>
+        <MediaCodec name="OMX.amlogic.dolby-vision.dvav.decoder.awesome2" type="video/dolby-vision" >
+            <Limit name="size" min="64x64" max="3840x2160" />
+            <Limit name="alignment" value="2x2" />
+            <Limit name="block-size" value="16x16" />
+            <Limit name="blocks-per-second" min="1" max="1944000" />
+            <Limit name="bitrate" range="1-300000000" />
+            <Feature name="adaptive-playback" />
+            <Feature name="tunneled-playback" />
+            <Limit name="concurrent-instances" max="1" />
+        </MediaCodec>
+        <MediaCodec name="OMX.amlogic.dolby-vision.dvav.decoder.awesome2.secure" type="video/dolby-vision" >
+            <Limit name="size" min="64x64" max="3840x2160" />
+            <Limit name="alignment" value="2x2" />
+            <Limit name="block-size" value="16x16" />
+            <Limit name="blocks-per-second" min="1" max="1944000" />
+            <Limit name="bitrate" range="1-300000000" />
+            <Feature name="adaptive-playback" />
+            <Feature name="secure-playback" required="true" />
+            <Feature name="tunneled-playback" />
+            <Limit name="concurrent-instances" max="1" />
+        </MediaCodec>
+        <MediaCodec name="OMX.amlogic.dolby-vision.dav1.decoder.awesome2" type="video/dolby-vision" >
+            <Limit name="size" min="64x64" max="3840x2160" />
+            <Limit name="alignment" value="2x2" />
+            <Limit name="block-size" value="16x16" />
+            <Limit name="blocks-per-second" min="1" max="1944000" />
+            <Limit name="bitrate" range="1-300000000" />
+            <Feature name="adaptive-playback" />
+            <Feature name="tunneled-playback" />
+            <Limit name="concurrent-instances" max="1" />
+        </MediaCodec>
+        <MediaCodec name="OMX.amlogic.dolby-vision.dav1.decoder.awesome2.secure" type="video/dolby-vision" >
+            <Limit name="size" min="64x64" max="3840x2160" />
+            <Limit name="alignment" value="2x2" />
+            <Limit name="block-size" value="16x16" />
+            <Limit name="blocks-per-second" min="1" max="1944000" />
+            <Limit name="bitrate" range="1-300000000" />
+            <Feature name="adaptive-playback" />
+            <Feature name="secure-playback" required="true" />
+            <Feature name="tunneled-playback" />
+            <Limit name="concurrent-instances" max="1" />
+        </MediaCodec>
         <MediaCodec name="OMX.amlogic.hevc.decoder.awesome2" type="video/hevc" >
```

1.7. How to Build Code

Build Bootloader

1. ohm(S905X4)

```
cd bootloader/uboot-repo/
./mk sc2_ah212 --avb2 --vab
cp build/u-boot.bin.signed ../../device/amlogic/ohm/bootloader.img
cp build/u-boot.bin.sd.bin.signed ../../device/amlogic/ohm/upgrade/
cp build/u-boot.bin.usb.signed ../../device/amlogic/ohm/upgrade/
```

2. ohmcas(S905C2)

```
cd bootloader/uboot-repo/
./mk sc2_ah232 --avb2 --vab
cp build/u-boot.bin.signed ../../device/amlogic/ohmcas/bootloader.img
cp build/u-boot.bin.sd.bin.signed ../../device/amlogic/ohmcas/upgrade/
cp build/u-boot.bin.usb.signed ../../device/amlogic/ohmcas/upgrade/
```

3. oppen(S905Y4)

```
cd bootloader/uboot-repo/
./mk s4_ap222 --avb2 --vab
cp build/u-boot.bin.signed ../../device/amlogic/oppen/bootloader.img
cp build/u-boot.bin.sd.bin.signed ../../device/amlogic/oppen/upgrade/
cp build/u-boot.bin.usb.signed ../../device/amlogic/oppen/upgrade/
```

4. oppencas(S905C3)

```
cd bootloader/uboot-repo/
./mk s4_ap232 --avb2 --vab
cp build/u-boot.bin.signed ../../device/amlogic/oppencas/bootloader.img
cp build/u-boot.bin.sd.bin.signed ../../device/amlogic/oppencas/upgrade/
cp build/u-boot.bin.usb.signed ../../device/amlogic/oppencas/upgrade/
```

5. planck(S805X2)

```
cd bootloader/uboot-repo/
./mk s4_aq222 --avb2 --vab
cp build/u-boot.bin.signed ../../device/amlogic/planck/bootloader.img
cp build/u-boot.bin.sd.bin.signed ../../device/amlogic/planck/upgrade/
cp build/u-boot.bin.usb.signed ../../device/amlogic/planck/upgrade/
```

Build Kernel

1. ohm(S905X4)

```
./mk ohm -v 5.4 -t user
```

2. ohmcas(S905C2)

```
./mk ohmcas -v 5.4 -t user
```

3. oppen(S905Y4)

```
./mk oppen -v 5.4 -t user
```

4. oppencas(S905C3)

```
./mk oppencas -v 5.4 -t user
```

5. planck(S805X2)

```
export KERNEL_A32_SUPPORT=true
./mk planck -v 5.4 -t user
```

Build Android

1. ohm(S905X4)

```
. build/envsetup.sh
lunch ohm-user
make -j8
```

2. ohmcas(S905C2)

```
. build/envsetup.sh
lunch ohmcas-user
make -j8
```

3. oppen(S905Y4)

```
. build/envsetup.sh
lunch oppen-user
make -j8
```

4. oppencas(S905C3)

```
. build/envsetup.sh
lunch oppencas-user
make -j8
```

5. planck(S805X2)

```
. build/envsetup.sh
export   KERNEL_A32_SUPPORT=true
lunch planck-user
make -j8
```

Noticed: Android R will start to use kernel 5.4 and we setup a new kernel and modules build way named standalone build.

Standalone build is a standard kernel and modules build way. We need separate Android and Kernel build.

1.8. Device Vendor SCS(Secure boot v3)

Device Vendor SCS(Secure boot v3), please refer the documents.

- <Amlogic S905X4 DeviceVendor SCS SigningProcess User Guide>
- <Amlogic S905X4 DIFPassword GenerationUser Guide>
- <Amlogic S905X4 USBPassword GenerationUser Guide>
- <Amlogic S905X4 Firmware Anti-rollback User Guide>
- <Amlogic S905X4 EFUSEBurning Provision UserGuide>
- <Amlogic S905Y4 DeviceVendor SCS SigningProcess User Guide>
- <Amlogic S905Y4 DIFPassword GenerationUser Guide>
- <Amlogic S905Y4 USBPassword GenerationUser Guide>
- <Amlogic Linux SCS Signing ToolUserGuide>

1.9. Secure Mode

This document provides required security mode configuration guidelines to prevent system data from being tampering with. Please refer the documents:
<Security Mode Configuration Guide>

1.10. ExoPlayer Note

ExoPlayer is an open source project that is not part of the Android framework and is distributed separately from the Android SDK. ExoPlayer's standard audio and video components are built on Android's MediaCodec API, which was released in Android 4.1 (API level 16).

1.11. Dynamic fingerprint

Although Android TV devices are similar, their system properties can differ. To help reduce the build load, the Android TV OEM partition has been released for use with Android TV on Android 11.

```
1. Change device/amlogic/oppen/oem/oem.prop
. build/envsetup.sh
lunch oppen-user
make custom_images -j8

2. copy out/target/product/oppen/oem.img to device/amlogic/oppen/oem/

As ohm.
```

1.12. Fastboot

We MUST update adb/fastboot version to 29.0.1-5644136 or 29.0.1-5644136+. Otherwise there are lots of adb/flash issues.

adb:

```
Android Debug Bridge version 1.0.41
Version 29.0.1-5644136
```

fastboot

```
fastboot version 29.0.1-5644136
```

You can download latest SDK Platform-Tools from
https://developer.android.com/studio/releases/platform-tools.html#downloads

1.13. How to Upgrade

There are 3 ways for update.

- Upgrade with USB burn tool
- Upgrade with OTA
- Upgrade with fastboot

1.13.1. all partitions

If you like to use our usb burn tool to update the images, it's the same as before.We have a fastboot flash-all script to update image, just like Android R. You need not care about the dynamic partitions.

Partition name	make command	imgae name	burning ways
all partitions	make otapackage	1.ohm-ota-xxxxxxxx.zip 2.aml_upgrade_package.img 3.ohm-fastboot-xxxxxxxx.zip	1.by OTA upgrade 2.by USB burning 3.by fastboot
	make	aml_upgrade_package.img	by USB burning

1.13.2. physical partitions

In Android R, physical partitions is bootloader, boot, logo, vendor_boot etc.
Fastboot should be in bootloader mode when flash these physical partitions.
Below is the way of burning each partitions:

Partition name	make command	imgae name	burning ways
boot	make bootimage	boot.img	fastboot flash boot boot.img
logo	make logoimg	logo.img	fastboot flash logo logo.img
uboot	./mk sc2_ah212 --avb2 --vab	u-boot.bin.signed	fastboot flash bootloader u-boot.bin.signed
odm_ext	make odm_ext_image	odm_ext.img	fastboot flash odm_ext odm_ext.img
vendor_boot	make vendorbootimage	vendor_boot.img	fastboot flash vendor_boot vendor_boot.img

Notes: execute ‘fastboot flashing unlock’ to unlock the device if wants to use fastboot commands.

1.13.3. logical partitions

In Android R, logical partitions is system, system_ext,vendor, odm, product.
Fastboot should be in fastbootd mode when flash these logical partitions.
Below is the way of burning each partitions:

Partition name	make command	image name	burning ways
system	make systemimage	system.img	fastboot flash system system.img
vendor	make vendorimage	vendor.img	fastboot flash vendor vendor.img
odm	make odmimage	odm.img	fastboot flash odm odm.img
product	make productimage	product.img	fastboot flash product product.img
system_ext	make systemextimage	system_ext.img	fastboot flash system_ext system_ext.img

Notes: execute ‘fastboot flashing unlock’ to unlock the device if wants to use fastboot commands.

2. Test Report

2.1. Auto Reboot in Normal Temperature

Test Platform	Test Case	Total test time	Test Result
AH212#175	Auto reboot	72H	Pass
AH212#194	Auto reboot	72H	Pass
AH212#200	Auto reboot	72H	Pass
AH212#318	Auto reboot	72H	Pass
AP222#157	Auto reboot	72H	Pass
AP222#181	Auto reboot	72H	Pass
AP222#188	Auto reboot	72H	Pass
AP222#289	Auto reboot	72H	Pass
AP232#130	Auto reboot	72H	Pass
AP232#131	Auto reboot	72H	Pass
AP232#134	Auto reboot	72H	Pass
AP232#135	Auto reboot	72H	Pass
AH212#175	uboot autoreboot	72H	Pass
AH212#194	uboot autoreboot	72H	Pass
AH212#200	uboot autoreboot	72H	Pass
AH212#318	uboot autoreboot	72H	Pass
AP222#157	uboot autoreboot	72H	Pass
AP222#181	uboot autoreboot	72H	Pass
AP222#188	uboot autoreboot	72H	Pass
AP222#289	uboot autoreboot	72H	Pass

2.2. Auto Shutdown/Suspend in Normal Temperature

Test Platform	Test Case	Total test time	Test Result
AH212#175	Autos hutdown	72H	Pass
AH212#194	Auto shutdown	72H	Pass
AH212#200	Auto shutdown	72H	Pass
AH212#318	Auto shutdown	72H	Pass
AH212#175	Auto shutdown	72H	Pass
AP222#137	Auto shutdown	72H	Pass
AP222#200	Auto shutdown	72H	Pass
AP222#224	Auto shutdown	72H	Pass
AP222#229	Auto shutdown	72H	Pass
AH212#175	Auto suspend	72H	Pass
AH212#194	Auto suspend	72H	Pass
AH212#200	Auto suspend	72H	Pass
AH212#318	Auto suspend	72H	Pass
AP222#137	Auto suspend	72H	Pass
AP222#200	Auto suspend	72H	Pass
AP222#224	Auto suspend	72H	Pass
AP222#229	Auto suspend	72H	Pass

2.3. 3D Game burning Test Report

Test Platform	Test Case	Total test time	Test Result
AH212#175	3D Game	72H	Pass
AH212#194	3D Game	72H	Pass
AH212#200	3D Game	72H	Pass
AH212#318	3D Game	72H	Pass
AP222#137	3D Game	72H	Pass
AP222#200	3D Game	72H	Pass
AP222#224	3D Game	72H	Pass
AP222#229	3D Game	72H	Pass
AP232#130	3D Game	72H	Pass
AP232#131	3D Game	72H	Pass
AP232#134	3D Game	72H	Pass
AP232#135	3D Game	72H	Pass

2.4. Monkey test Test Report

Test Platform	Test Case	Total test time	Test Result	Comments
AH212#175	Monkey test/300ms	72H	Pass	
AH212#194	Monkey test/300ms	72H	Pass	
AH212#200	Monkey test/300ms	72H	Pass	
AH212#318	Monkey test/300ms	72H	Pass	
AP222#157	Monkey test/300ms	72H	Pass	
AP222#181	Monkey test/300ms	72H	Pass	
AP222#188	Monkey test/300ms	72H	Pass	
AP222#289	Monkey test/300ms	72H	Pass	
AP232#137	Monkey test/300ms	72H	Pass	
AP232#224	Monkey test/300ms	72H	Pass	
AP232#229	Monkey test/300ms	48H	Fail	subtitle pointer

Monkey Black list

com.droidlogic.PPPoEcom.droidlogic.PPPoE
com.droidlogic.miracast
com.droidlogic.mediacenter
com.droidlogic.otaupgrade
com.droidlogic.tvinput
com.hpplay.happyplay.aw
com.android.tv.settings
com.droidlogic.tv.settings
com.google.android.tungsten.setupwraith
com.google.android.gms
com.android.vending
com.google.android.youtube.tv
com.google.android.videos
com.google.android.tv.remote.service
com.google.android.tvrecommendations
com.google.android.gms.persistent
com.google.android.youtube.tv.recommendations
com.google.android.tv.frameworkpackagestubs
com.google.android.tvcom.droidlogic.videoplayer

2.5. FactoryReset Test Report

Test Platform	Test Case	Total test time	Test Result
AH212#175	Factory reset	72H	Pass
AH212#194	Factory reset	72H	Pass
AH212#200	Factory reset	72H	Pass
AH212#318	Factory reset	72H	Pass
AP232#136	Factory reset	72H	Pass
AP232#137	Factory reset	72H	Pass
AP232#139	Factory reset	72H	Pass
AP222#157	Factory reset	72H	Pass
AP222#181	Factory reset	72H	Pass
AP222#188	Factory reset	72H	Pass
AP222#289	Factory reset	72H	Pass

2.6. A/V Burn Test Report

Test Platform	Test Case	Total test time	Test Result
AH212#94	4K H265	72H	Pass
AH212#128	4K H265	72H	Pass
AH212#137	4K H265	72H	Pass
AH212#194	4K H265	72H	Pass
AP232#134	4K H265	72H	Pass
AH212#128	4K H264	72H	Pass
AH212#238	4K H264	72H	Pass
AP232#131	4K H264	72H	Pass
AH212#194	4K VP9	72H	Pass
AH212#249	4K VP9	72H	Pass
AP232#135	4K VP9	72H	Pass
AP232#130	4K AV1	72H	Pass
AH212#105	4K AV1	72H	Pass
AP222#176	4K AV1	72H	Pass
AP222#134	4K H265	72H	Pass
AP222#169	4K H264	72H	Pass
AP222#200	4K VP9	72H	Pass
AH212#120	Youtube	12H	Pass
AH212#194	Youtube	12H	Pass
AH212#120	Youtube	12H	Pass
AP222#99	Youtube	12H	Pass
AP222#188	full format	72H	Pass
AP222#159	full format	72H	Pass
AP222#169	full format	72H	Pass

2.7. High(70℃) Temperature Test Report

Test Platform	Test Case	Total test time	Test Result
AH212#1	Auto reboot	72H	Pass
AH212#2	Auto reboot	72H	Pass
AH212#3	Auto reboot	72H	Pass

AH212#4	Auto reboot	72H	Pass
AH212#5	Auto reboot	72H	Pass
AH212#1	3D Game	72H	Pass
AH212#2	3D Game	72H	Pass
AH212#3	3D Game	72H	Pass
AH212#4	3D Game	72H	Pass
AH212#5	3D Game	72H	Pass
AP222#101	Auto reboot	72H	Pass
AP222#102	Auto reboot	72H	Pass
AP222#103	Auto reboot	72H	Pass
AP222#104	Auto reboot	72H	Pass
AP222#106	Auto reboot	72H	Pass
AP222#101	3D Game	72H	Pass
AP222#102	3D Game	72H	Pass
AP222#103	3D Game	72H	Pass
AP222#104	3D Game	72H	Pass
AP222#106	3D Game	72H	Pass

2.8. Low(-20℃) Temperature Test Report

Test Platform	Test Case	Total test time	Test Result
AH212#1	Auto reboot	72H	Pass
AH212#2	Auto reboot	72H	Pass
AH212#3	Auto reboot	72H	Pass
AH212#4	Auto reboot	72H	Pass
AH212#5	Auto reboot	72H	Pass
AH212#1	3D Game	72H	Pass
AH212#2	3D Game	72H	Pass
AH212#3	3D Game	72H	Pass
AH212#4	3D Game	72H	Pass
AH212#5	3D Game	72H	Pass
AP222#101	Auto reboot	72H	Pass
AP222#102	Auto reboot	72H	Pass
AP222#103	Auto reboot	72H	Pass
AP222#104	Auto reboot	72H	Pass
AP222#106	Auto reboot	72H	Pass
AP222#101	3D Game	72H	Pass
AP222#102	3D Game	72H	Pass
AP222#103	3D Game	72H	Pass
AP222#104	3D Game	72H	Pass
AP222#106	3D Game	72H	Pass

2.9. DVB Stress Test

Test Platform	Test Case	Total test time	Test Result	Comments
AH212#166	Auto Search	72H	Pass	
AH212#232	Auto Search	72H	Pass	
AH212#126	Auto Search	72H	Pass	
AH212#169	Auto Search	72H	Pass	
AH212#166	4K Bitstream Longterm Playing	72H	**Fail**	Reboot

AH212#232	4K Bitstream Longterm Playing	72H	Pass	
AH212#126	4K Bitstream Longterm Playing	72H	Pass	
AH212#169	4K Bitstream Longterm Playing	72H	Pass	
AH212#166	Single freq. Channel Change	72H	Pass	
AH212#232	Single freq. Channel Change	72H	Pass	
AH212#126	Single freq. Channel Change	72H	Pass	
AH212#169	Single freq. Channel Change	72H	Pass	
AH212#169	Multi freq. Channel Change	72H	Pass	
AH212#166	Multi freq. Channel Change	72H	Pass	
AH212#232	Multi freq. Channel Change	72H	Pass	
AH212#126	Multi freq. Channel Change	72H	Pass	
AH212#169	Subtitle Longterm Playing	72H	Pass	
AH212#166	Subtitle Longterm Playing	72H	Pass	
AH212#232	Subtitle Longterm Playing	72H	Pass	
AH212#126	Subtitle Longterm Playing	72H	Pass	
AH212#1	Program Switch(FCC)	72H	Pass	
AH212#2	Program Switch(FCC)	72H	Pass	
AH212#1	LongTime Playback(PIP)	72H	Pass	
AH212#2	LongTime Playback(PIP)	72H	Pass	
AH212#1	Program Switch(PIP)	72H	**Fail**	video freeze
AH212#2	Program Switch(PIP)	72H	**Fail**	video freeze

2.10. *Video Format Test Report*

Extension	Codec Detail	Tested Resolution	Ohm/Ohmcas	Oppen/Oppencas
.3g2	H263	320x276	Support	Support
.3gp	MPEG-4 Visual	640x480	Support	Support
	MPEG-4	320x240	Support	Support
	H263	320x276	Support	Support
	H264	1920x1080	Support	Support
.avi	AVC	1920x1080	Support	Support
	M-JPEG	1024x576	Support	Support
	RealMagic MPEG-4	720x480	Support	Support
		720x576	Support	Support
	h264	1920x1080	Support	Support

	FF mpeg MPEG4	640x480	Support	Support
.f4v	AVC(H264)	1280x720	Support	Support
.flv	Sorenson Spark	1920x1080	Support	Support
	VP6	800x342	Support	Support
.mp4	AVC(H264)	1920x1080	Support	Support
	HEVC(H265)	1920x1080	Support	Support
	4K HEVC(4K H265)	4096x2304	Support	Support
	8K HEVC(8K H265)	8192x4320	No support	No support
	AV1	1920x1080	Support	Support
	4K AV1	3840x2160	Support	Support
	H263	176x144	Support	Support
.m2ts	AVC	1920x1080	Support	Support
	VC-1	1920x1080	Support	Support
.m2v	MPEG-2	480x576	Support	Support
.m4v	AVC	1280x720	Support	Support
.mkv	WMV3	1280x720	Support	Support
	MPEG-4 Visual	1920x1080	Support	Support
	AVC	1920x1080	Support	Support
	4K HEVC(4K H265)	3840x2160	Support	Support
	8K HEVC(8K H265)	8192x4320	No support	No support
	VP8	1920x1080	Support	Support
	VP9	1920x1080	Support	Support
.mov	MPEG-4 Visual	1280x720	Support	Support
	mjpa	640x480	Support	Support
	H263	320x240	Support	Support
	M-JPEG	640x480	Support	Support
	AVC(H264)	1920x1080	Support	Support
	4K H264	3840x2160	Support	Support
.mpeg	MPEG-2	1920x1080	Support	Support
	VC-1	1920x1080	Support	Support
	MPEG-1	720x576	Support	Support
.mts	AVC	1440x1080	Support	Support
.ogm	XVID	640x480	Support	Support
.PMP	H264	480x272	Support	Support
.tp	MPEG-2	1920x1088	Support	Support
.ts	HEVC(H265)	1920x1080	Support	Support
	4K HEVC(4K H265)	4096x2304	Support	Support
	8K HEVC(8K H265)	8192x4320	No support	No support
	MPEG-1	1920x1080	Support	Support
	avs	720x576	Support	Support
	avs+	1920x1080	Support	Support
	avs2	3840x2160	Support	Support
	AVC	1920x1080	Support	Support
	MPEG-2	1920x1080	Support	Support

.vob	MPEG-2	720x576	Support	Support
.wmv	WMV1	640x480	Support	Support
	WMV2	768x432	Support	Support
	WMV3	1920x1080	Support	Support
.webm	VP8	1920x1080	Support	Support
	VP9	1920x1080	Support	Support
	4K VP9	4096x2160	Support	Support
	8K VP9	8192x4608	No support	No support
	AV1	1920x1080	Support	Support
4K	AVC(H264)	4096x2304	Support	Support
	HEVC(H265)	4096x2304	Support	Support
	VP9	4096x2160	Support	Support
	AV1	3840x2160	Support	Support
8K	HEVC(H265)	8192x4320	No support	No support
	VP9	8192x4608	No support	No support

Noticed: Please confirm the license before opening the multimedia format.

3. Tools version

Tool name	version
USB burning tool	**Aml_Burn_Tool_V3.2.0**
Customization	Amlogic customizationl V.2.1.3
Linux Secure boot V3	Aml_Linux_SCS_SignTool
adb	1.0.41
fastboot	29.0.1
Kernel tool chain	gcc-linaro-6.3.1-2017.02-x86_64_arm-linux-gnueabihf gcc-linaro-6.3.1-2017.02-x86_64_aarch64-linux-gnu

4. License version

Licensed Feature Name	Version	Document version
Dolby Digital Plus for Android	v20201021	In firmware Readme text
Dolby Digital for Android	v20200723	In firmware Readme text
Dolby MS12 V2.4 for android R	v20211015	In firmware Readme text
DTS-HD for android R	v20211026	In firmware Readme text
DRA Decoder for android	v20210728	In firmware Readme text
Secure OS TDK	Tdk is included in the code	Amlogic TDK Integration User Guide V3.2 Amlogic Secure Key Provision User Guide V2.6
Widevine	L1/L3	Amlogic WideVine Integration User Guide v0.9
Playready		Amlogic PlayReady Integrate User Guide v1.32
Dolby Vision 2.5 for Android kernel64bit	Ohm dovi.ko stb_release_20210330 dovi_fw_u2019.bin release-20210413 Oppen dovi.ko stb_release_20210513 dovi_fw_u2019.bin release-20210413	In firmware Readme text
HDCP 1.4 for HDMI TX for Android	v20180408	Amlogic_HDCP1.4_Tx_Release_Note_V0.1
HDCP 2.X for HDMI TX for Android	v20201212	Amlogic_HDCP2.2_Tx_Release_Note_V1.0
Verimatrix	vendor/vmx/bootloader(m9x4-rel: 98702f24f7c9c46a32b253a7bd0a8d7 f73bcc7d9) vendor/vmx/bootloader(m9y4-rel: 396b789a8c9e51cbbfd8586d39fcd3d d2e4dce32) vendor/amlogic/prebuild/verimatrix(r-a mlogic: 9584597e7b7e03372a82a9ec461d68 fc7c59a5f4) vmx/sdk-rel(m9x4-rel: be555e234f5e86763b55e34aaf00a96 cf95da757) vmx/sdk-rel(m9y4-rel: ea3f467a4f2c2d82b3449f6242e1346 18e1ad8f7)	OHM Amlogic CAS - S905X4 Verimatrix SDK Integration User Guide v2.5 Amlogic CAS - S905X4 Verimatrix Client Integration User Guide v5.0 Amlogic CAS - Verimatrix Adaption API Guide-v1.0.pdf Oppen Amlogic CAS - S905Y4 Verimatrix SDK Integration User Guide v1.5 Amlogic-CAS - S905Y4-Verimatrix-Client-Integration-User-Guide-v7.0 Amlogic CAS - S805X2 Verimatrix SDK Integration User Guide v8.0 Amlogic CAS - Verimatrix Adaption API Guide-v1.0.pdf

5. License path

Licensed Files	Path
HDCP TX firmware.le	/odm/etc/firmware/firmware.le
Dolby	/odm/lib/libHwAudio_dcvdec.so
MS12	/odm/etc/ms12/libdolbyms12.so
Dolby vision	/odm/lib/modules/dovi.ko /mnt/vendor/odm_ext/firmware/dovi_fw.bin